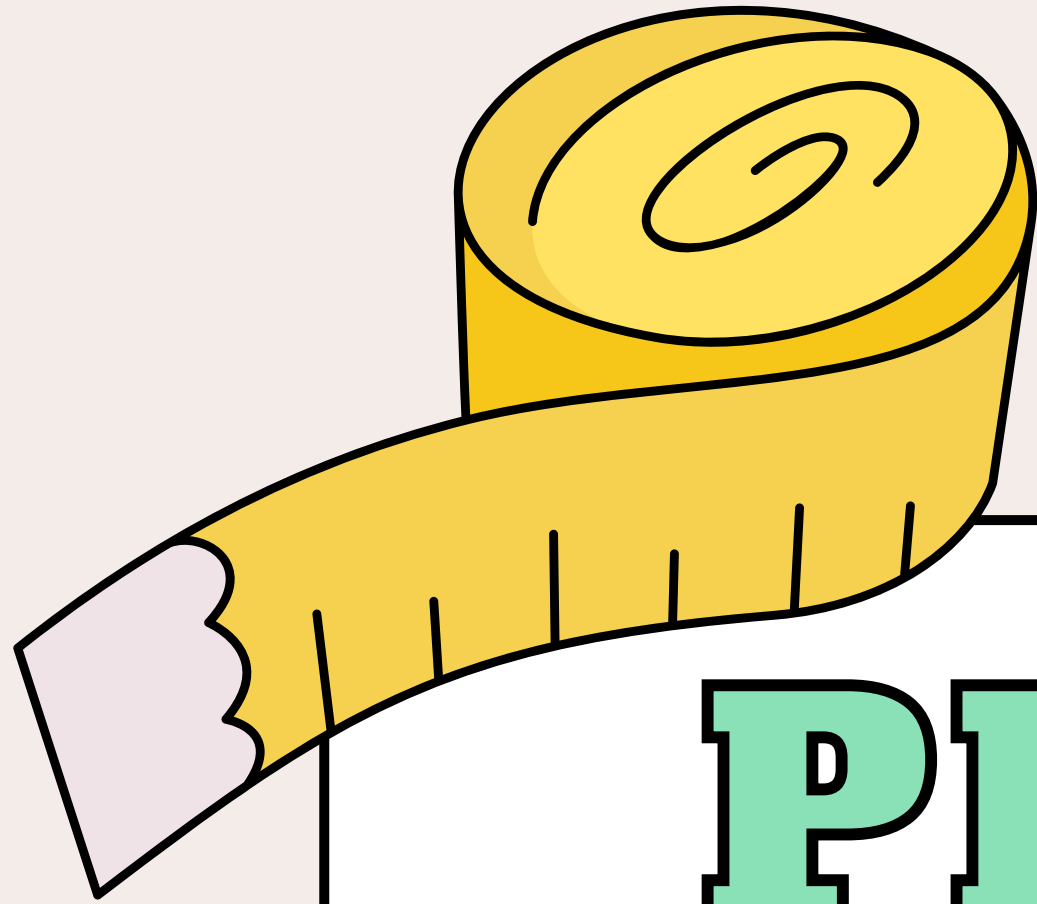
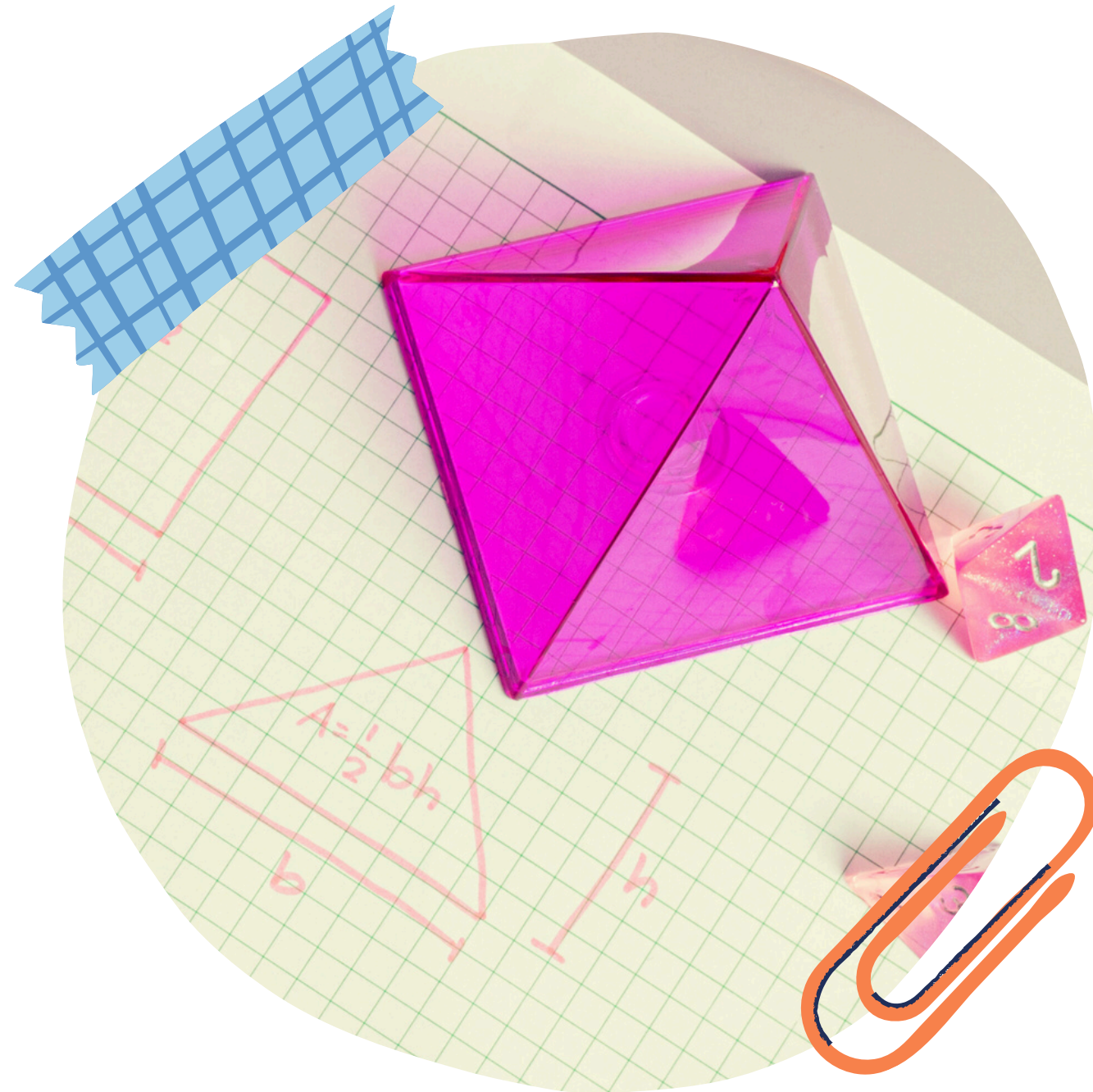




PERIMETER AND AREA

GRADE 4

Quick recall!



Which one is a rectangle?

Shape with 4 equal sides

Shape with 3 sides, where 2 sides are equal

Shape with 4 equal sides

Shape with 4 sides, where the opposite sides are equal

Your name will not be shared | 0 votes

Tip: Double-click to customize this poll or quiz,
or go to the **Elements Tab** for more options!

Lesson Outline

- Area of a rectangle on a Grid
- Area vs. Perimeter
- Solving for the Area and Perimeter



GOOD LUCK!



Learning Outcomes

- Determine the area of a rectangular figure on a grid
- Differentiate area and perimeter
- Solve word problems involving area and perimeter of a rectangle

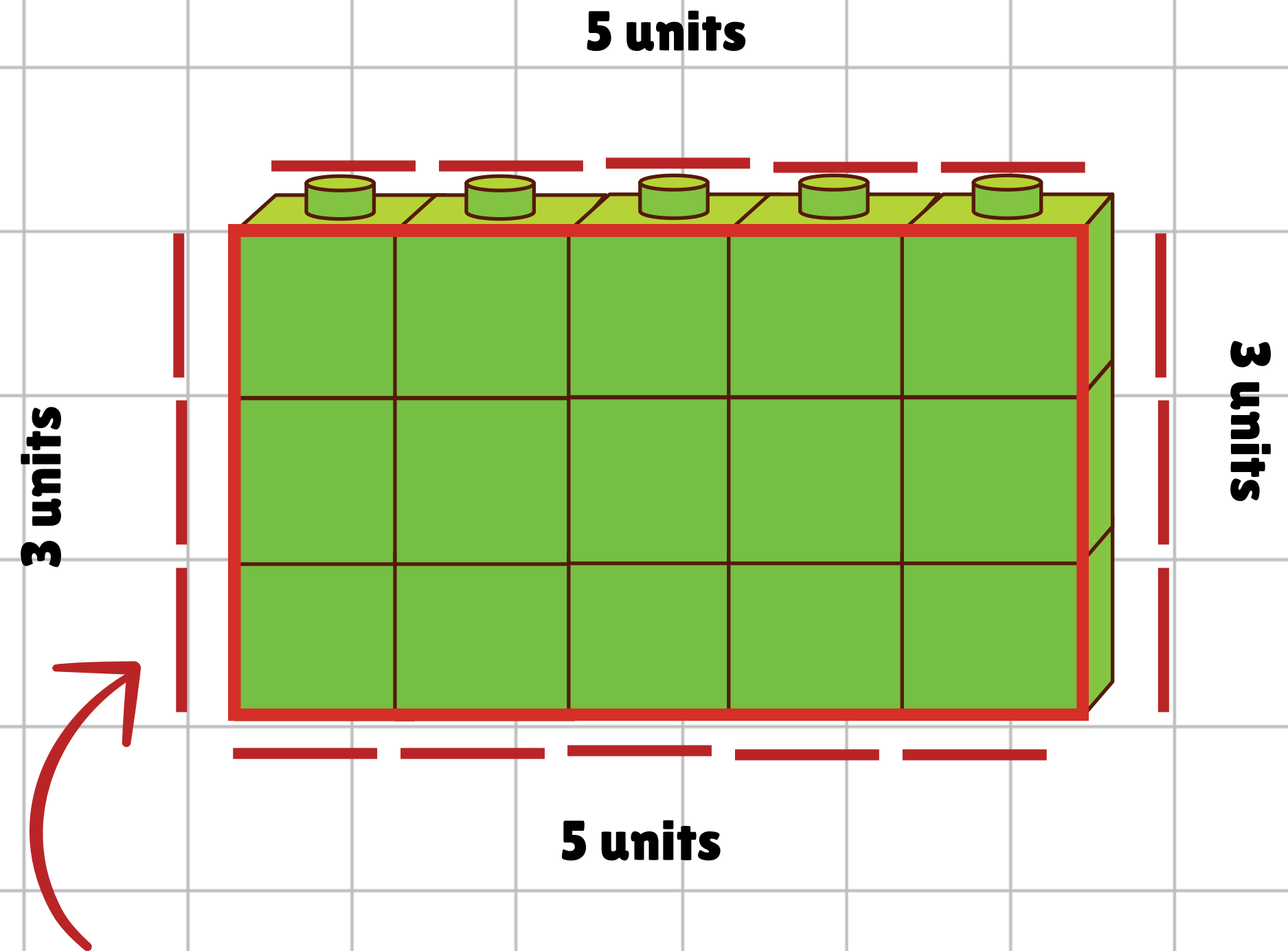


What is Perimeter?

Perimeter is the total distance around a shape.

Formula: Add up all the sides

Perimeter = side + side + side + side.

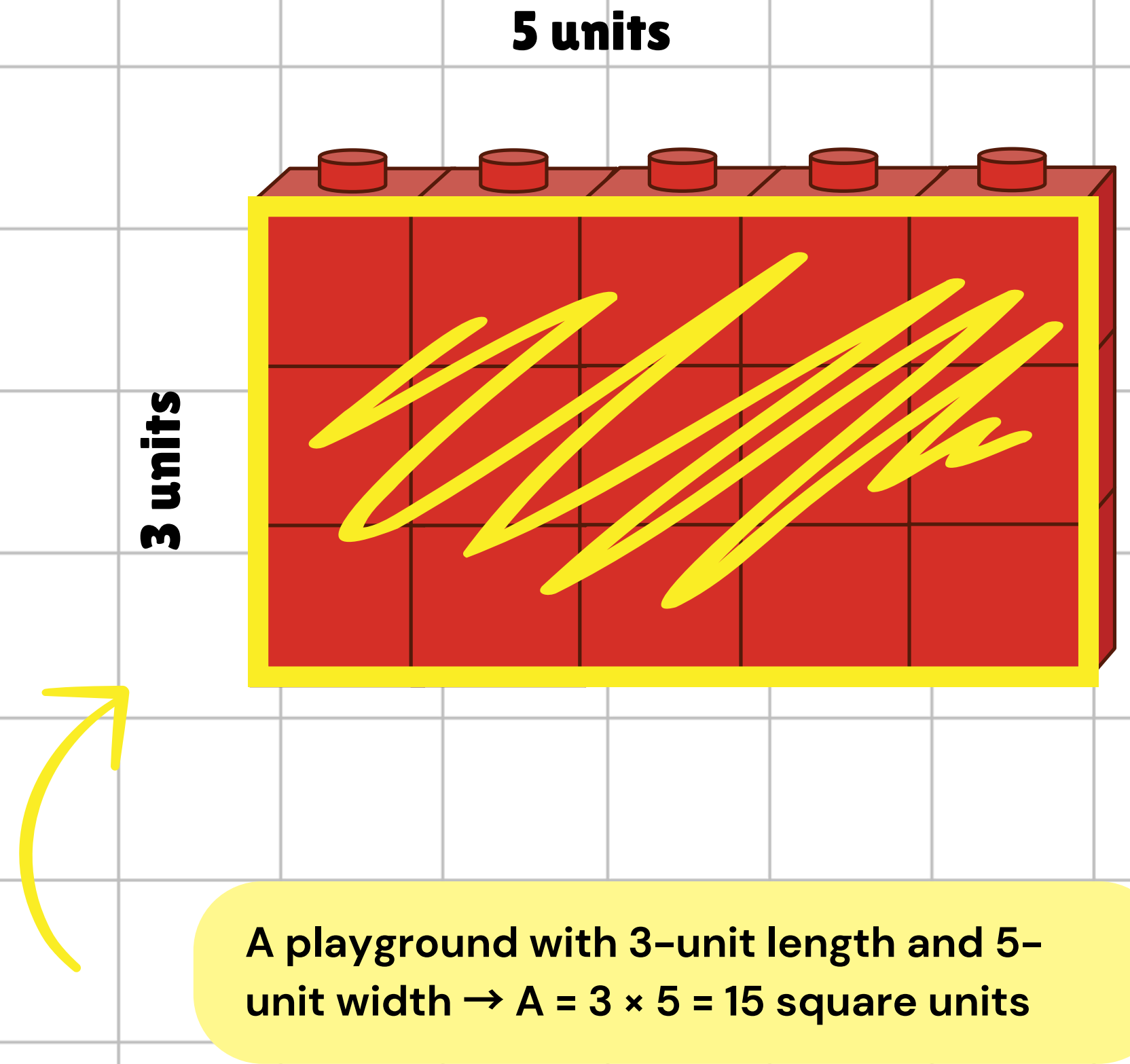


A park with 3-unit length and 5-unit width
Perimeter = $3 + 5 + 3 + 5 = 16$ units

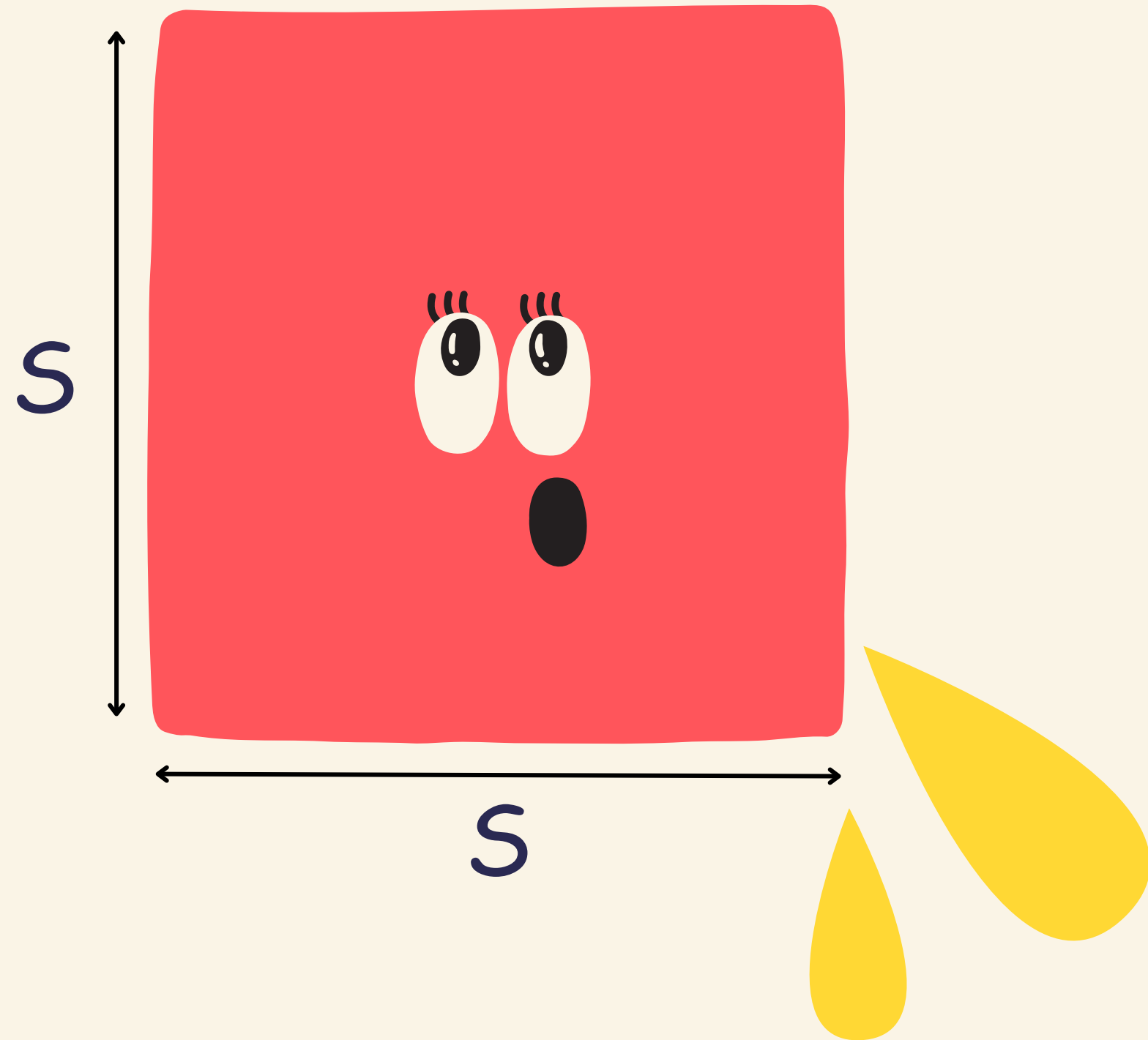
What is Area?

The total space inside a shape.

Formula: $A = \text{length} \times \text{width}$ for rectangles.



S Q U A R E



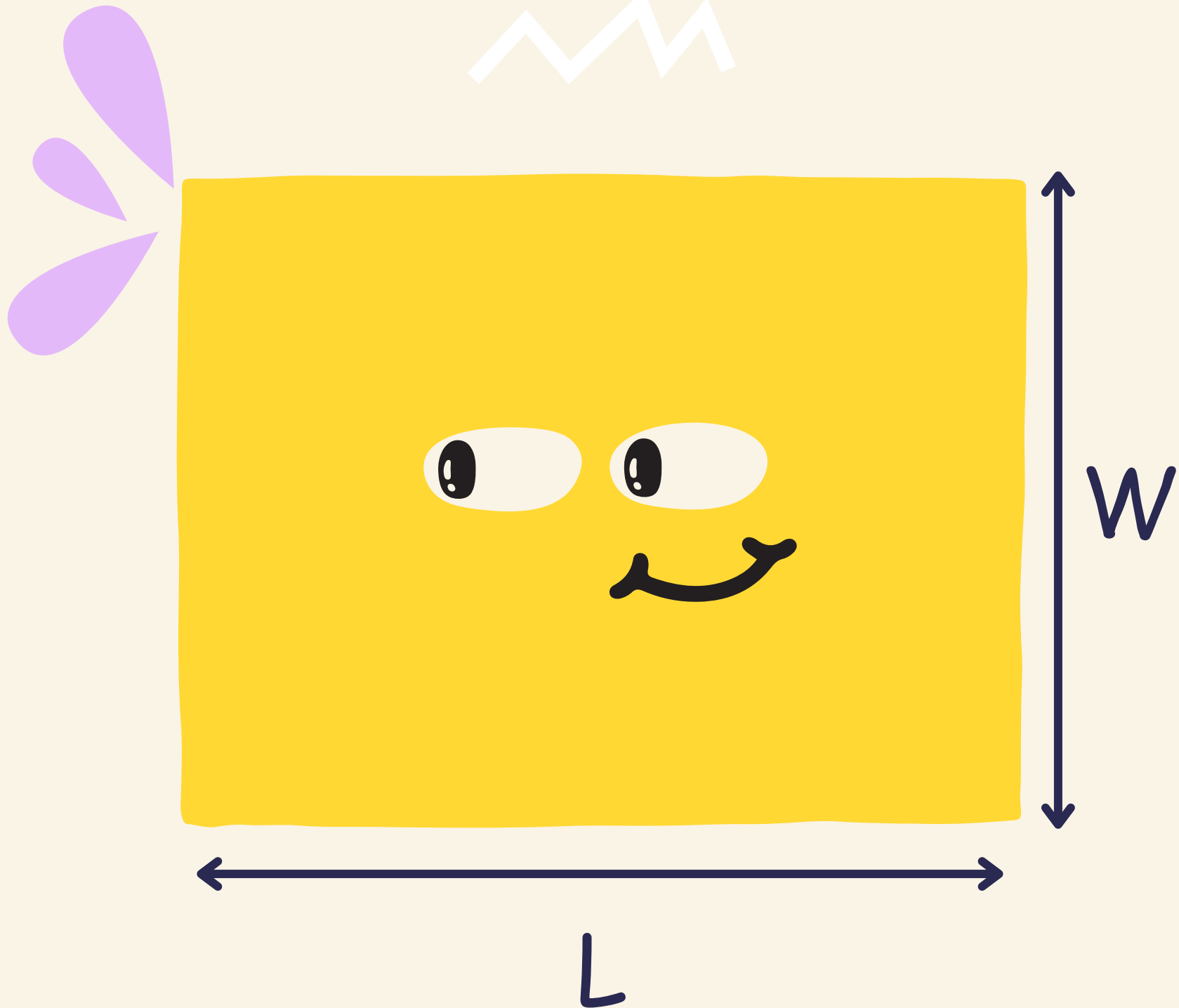
AREA:

$$S \times S$$

PERIMETER:

$$S + S + S + S$$

RECTANGLE



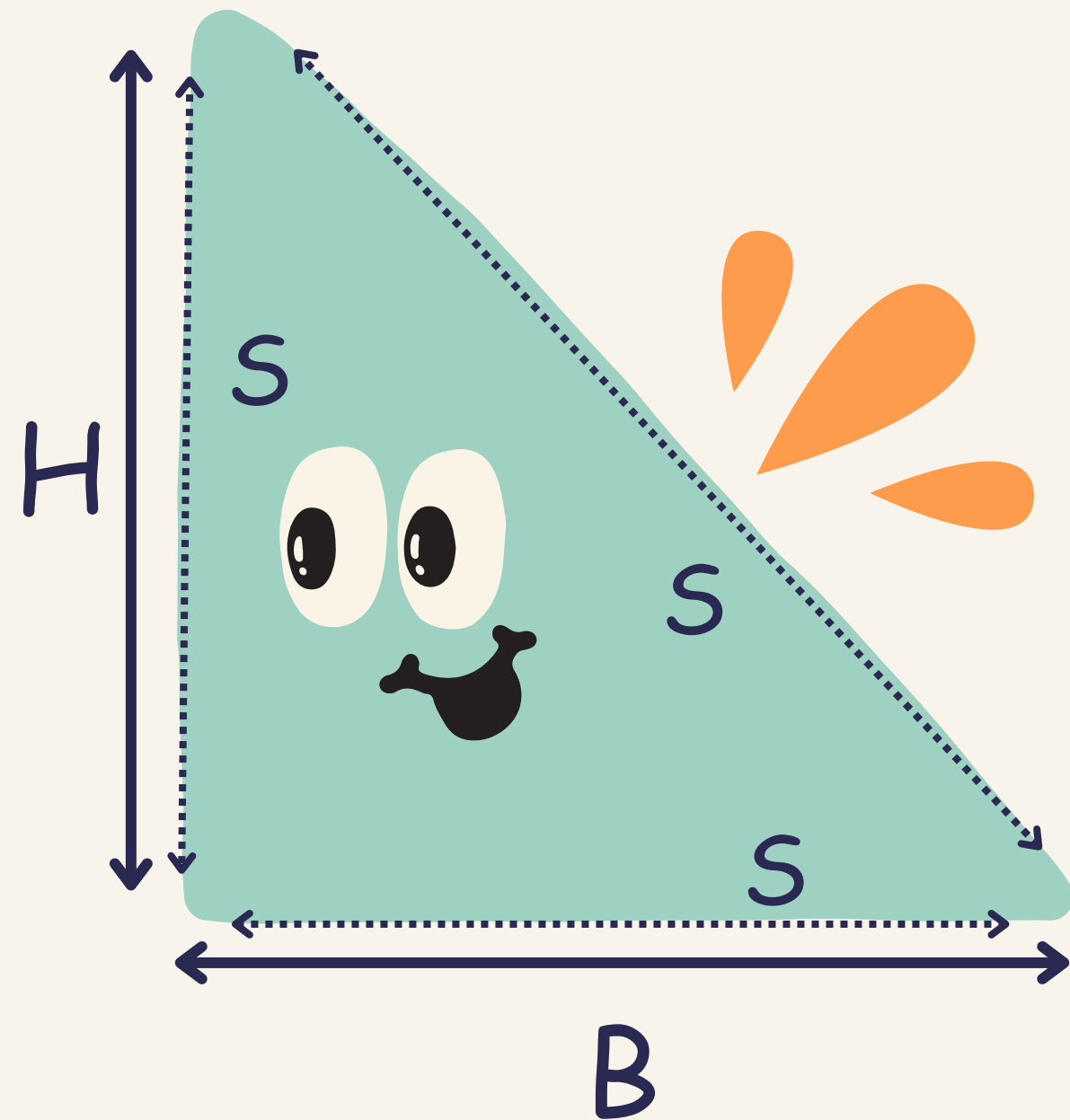
AREA:

$$L \times W$$

PERIMETER:

$$(L + W) \times 2$$

TRIANGLE



AREA:

$$\frac{1}{2} B \times H$$

PERIMETER:

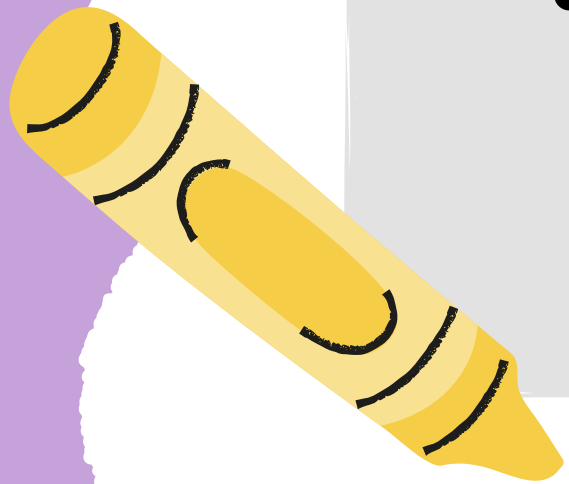
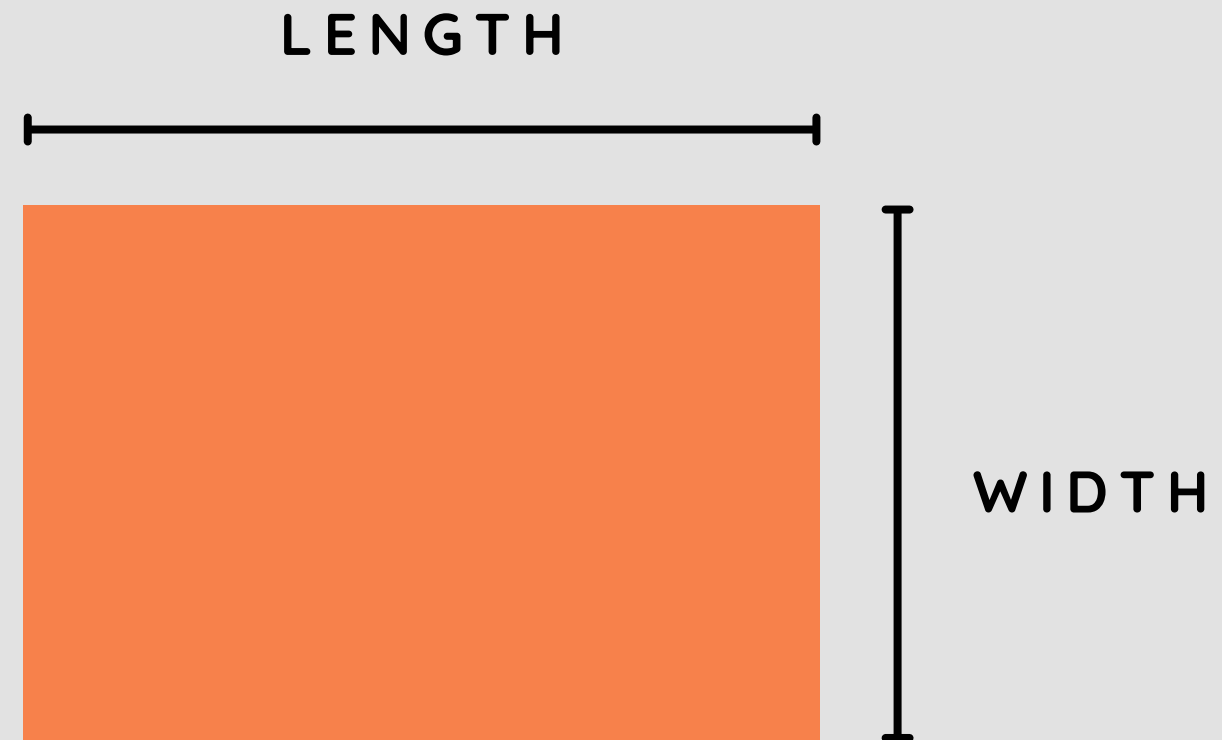
$$S + S + S$$



Rectangle



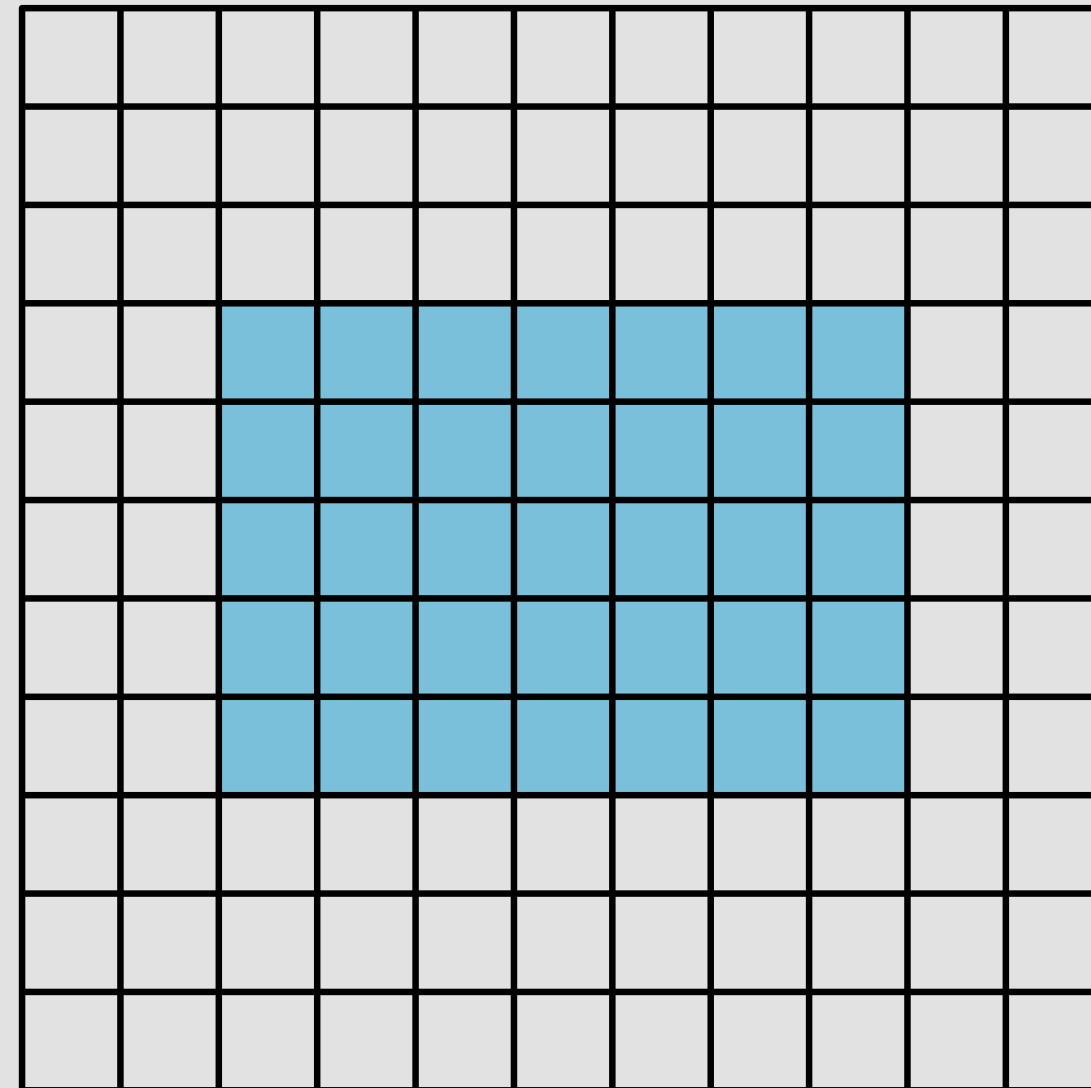
- Has 4 sides.
- Has 4 right angles.
- Parallel sides have equal measures.
- The sides of a rectangle are known as length and width.



Area on a Grid

A grid has vertical and horizontal lines that form a square. Each square measures 1 sq. unit.

Let's count how many squares the blue rectangle covers.

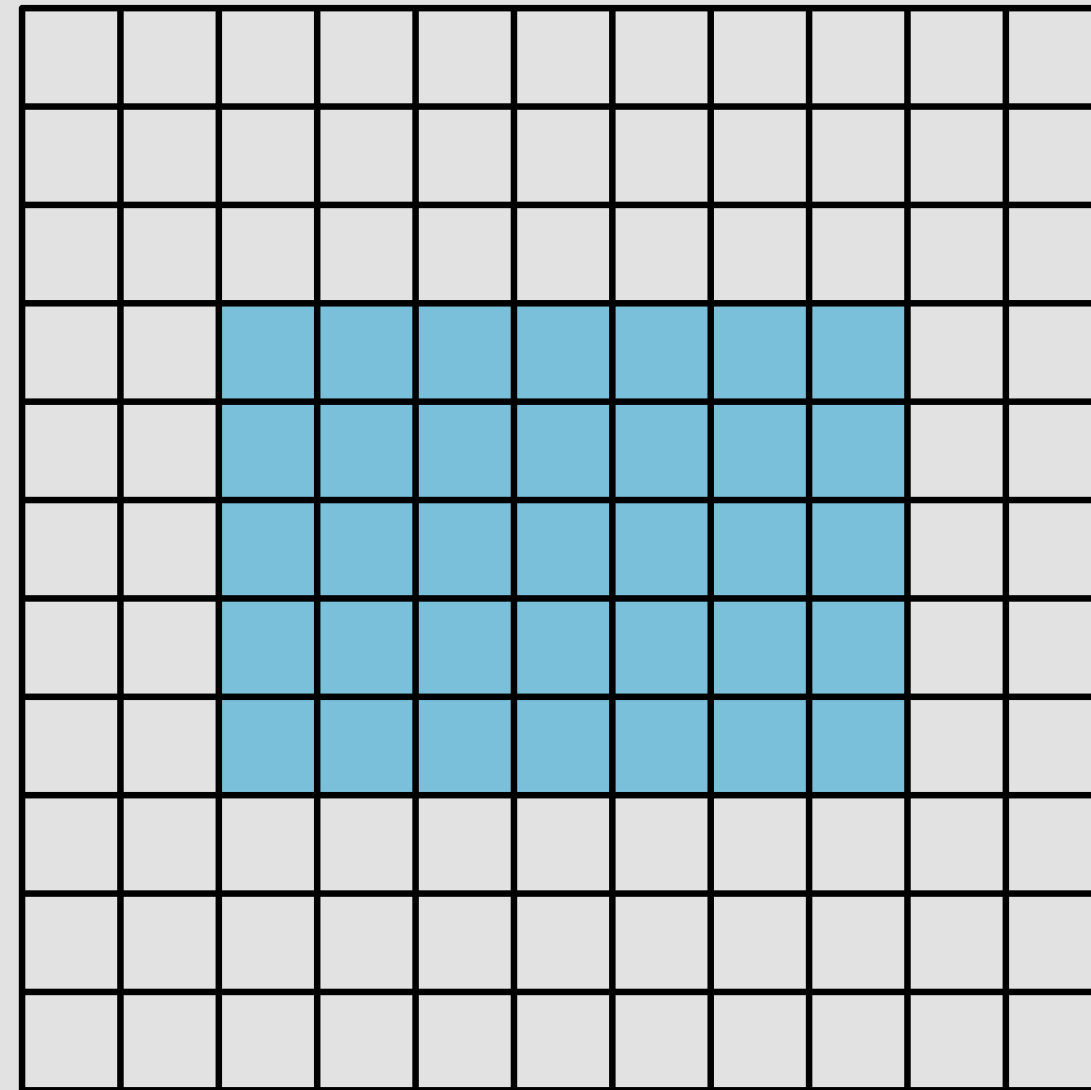


Area on a Grid

There are 35 squares.

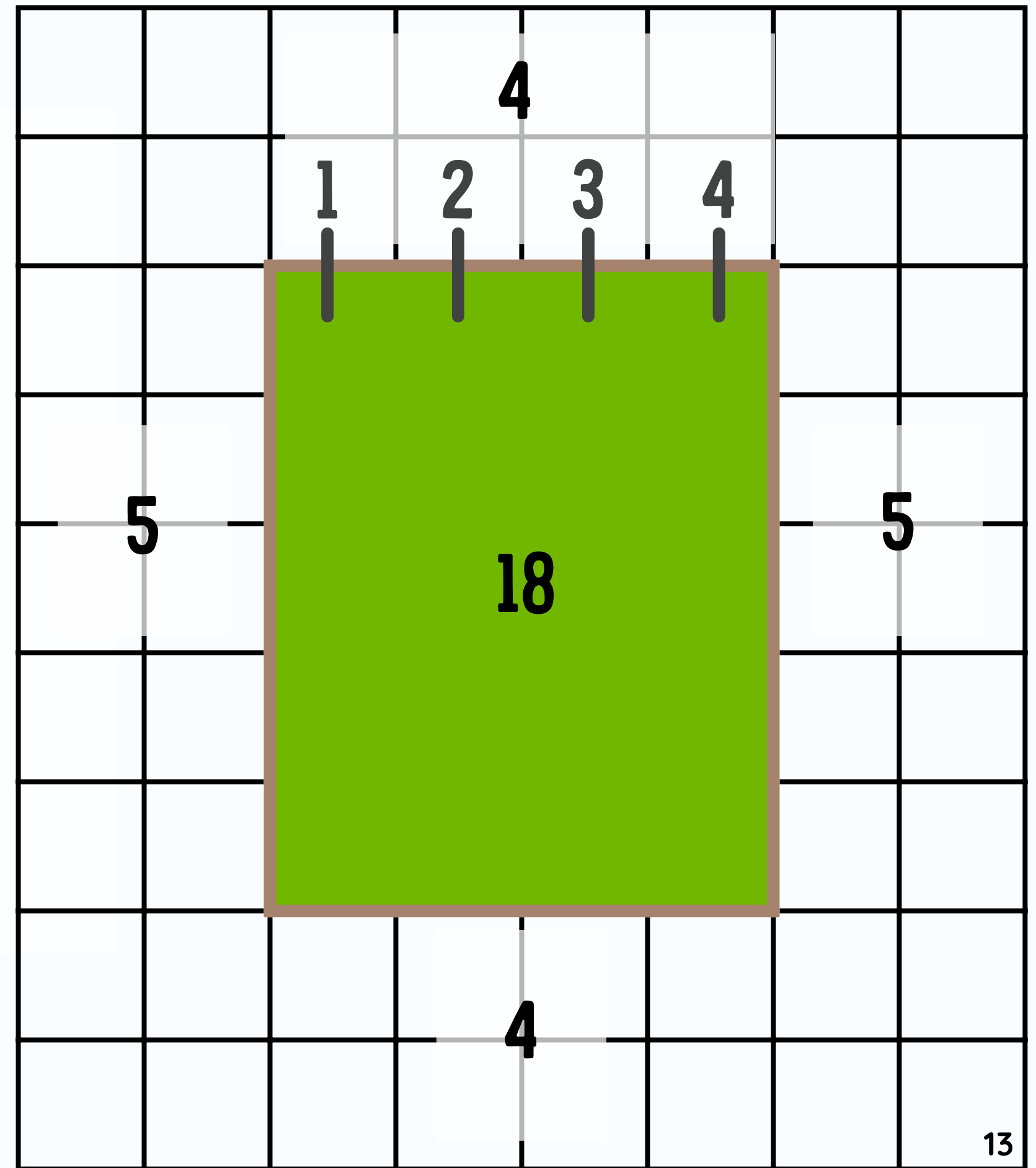
This means that the rectangle
has an area of 35 square units.

35 SQUARE UNITS

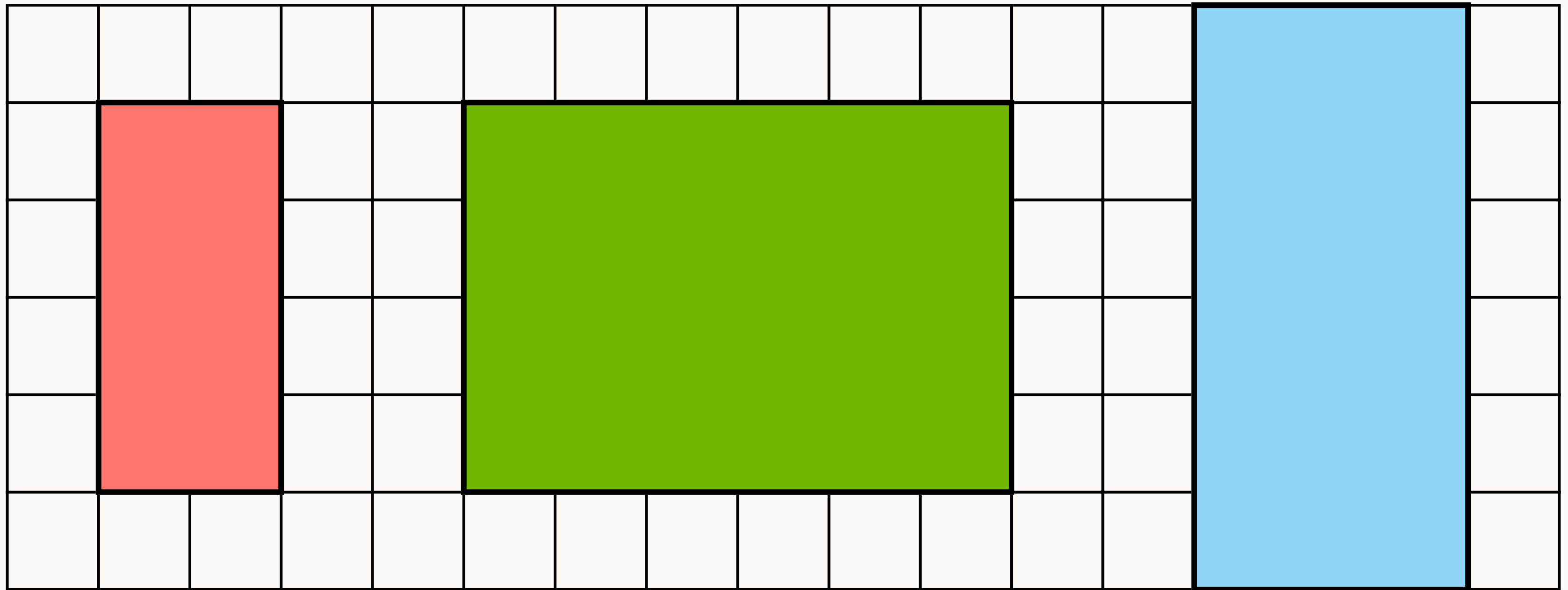


Perimeter on a grid

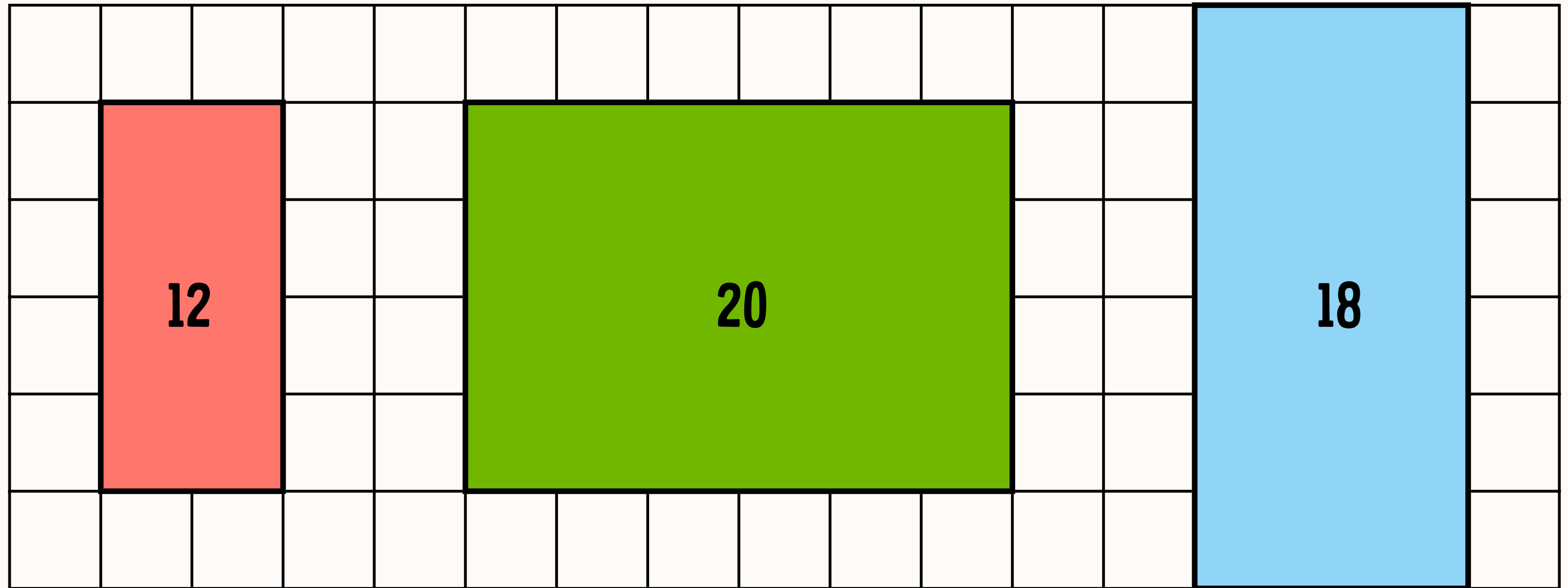
When a shape is presented on a grid without side lengths listed, we can count the unit squares on each side to determine the side lengths.



What is the Perimeter of These Figures?



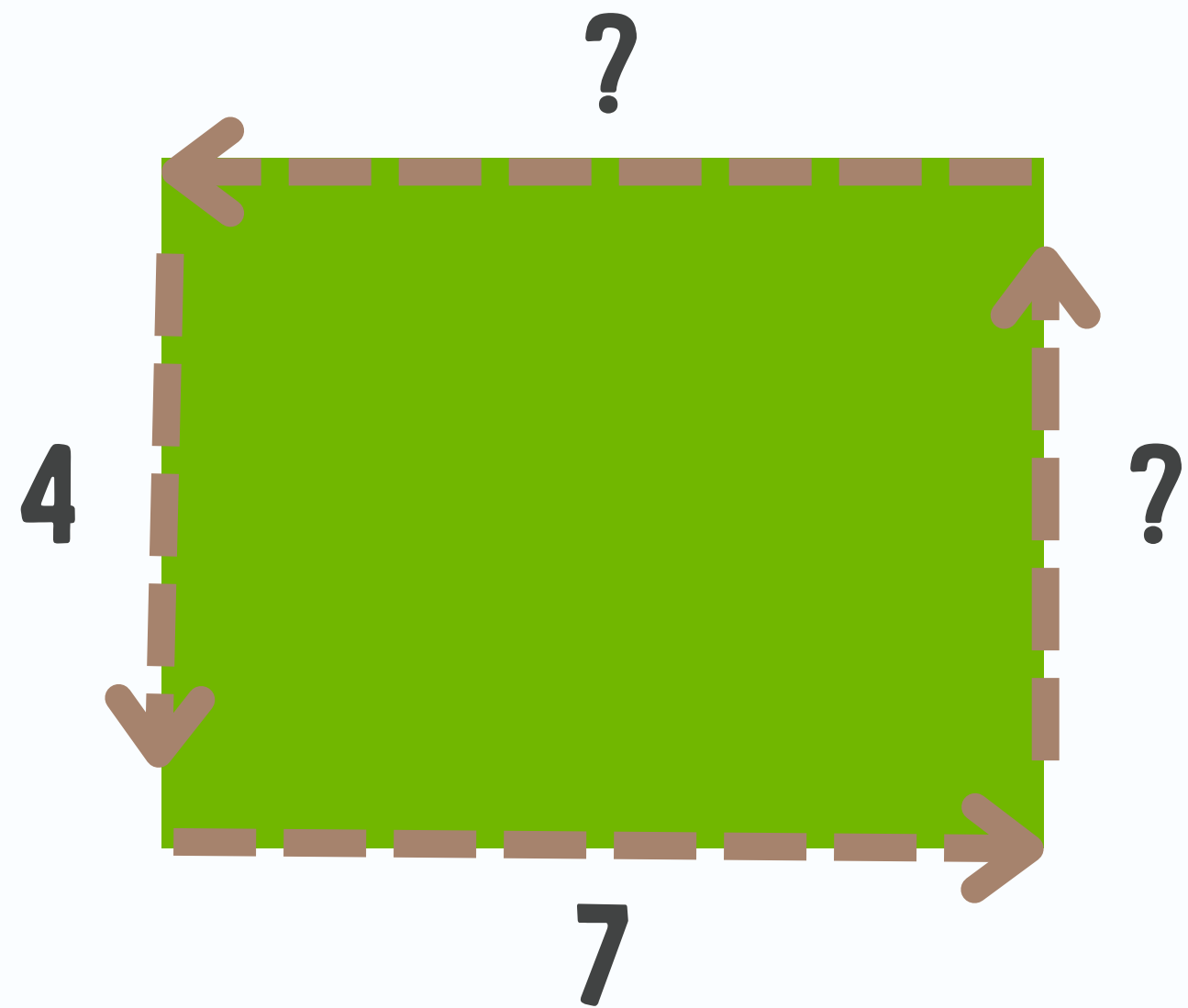
ANSWERS:



$$4 + 4 + 2 + 2 = 12$$

$$6 + 6 + 4 + 4 = 20$$

$$3 + 3 + 6 + 6 = 18$$

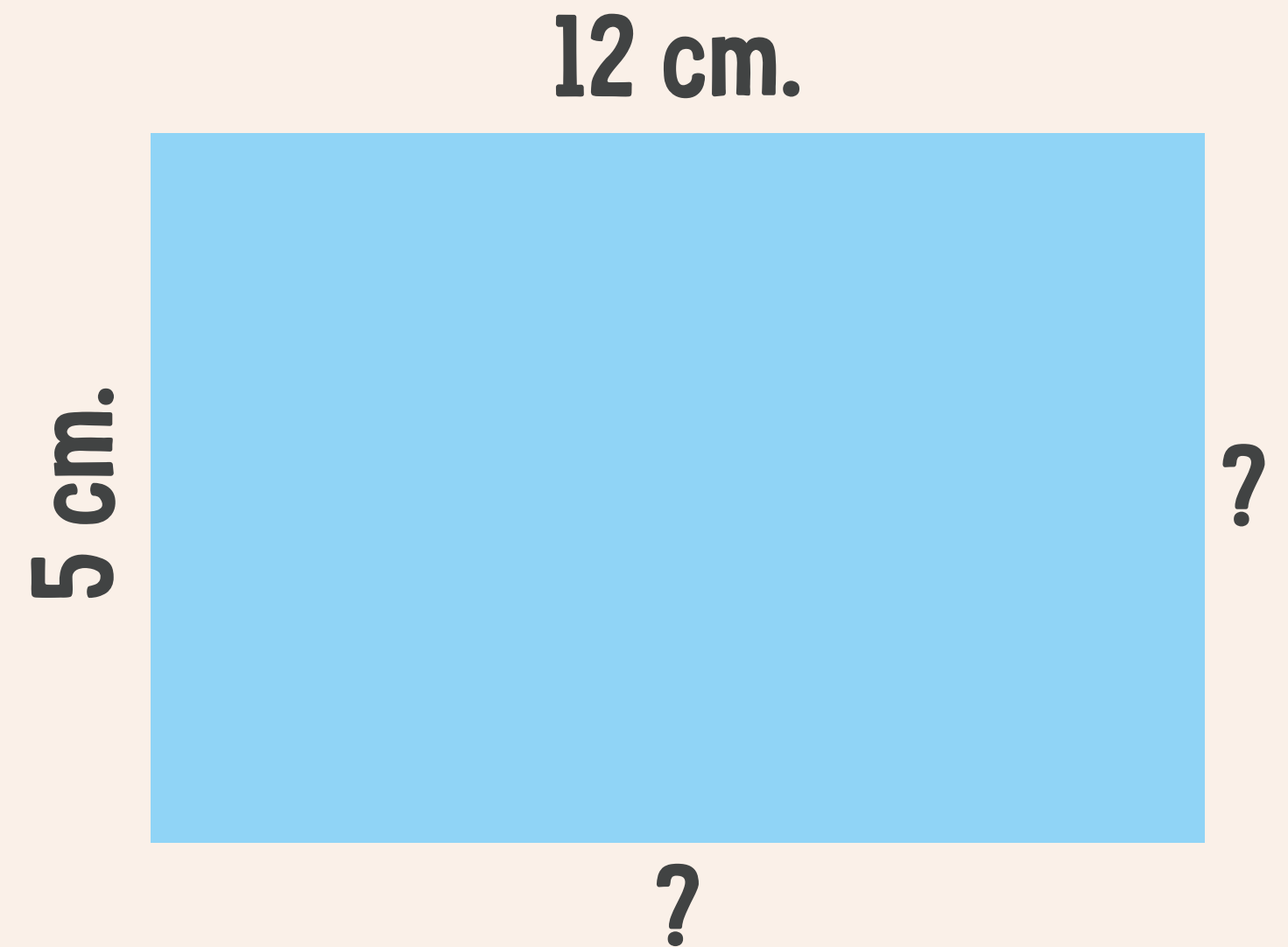
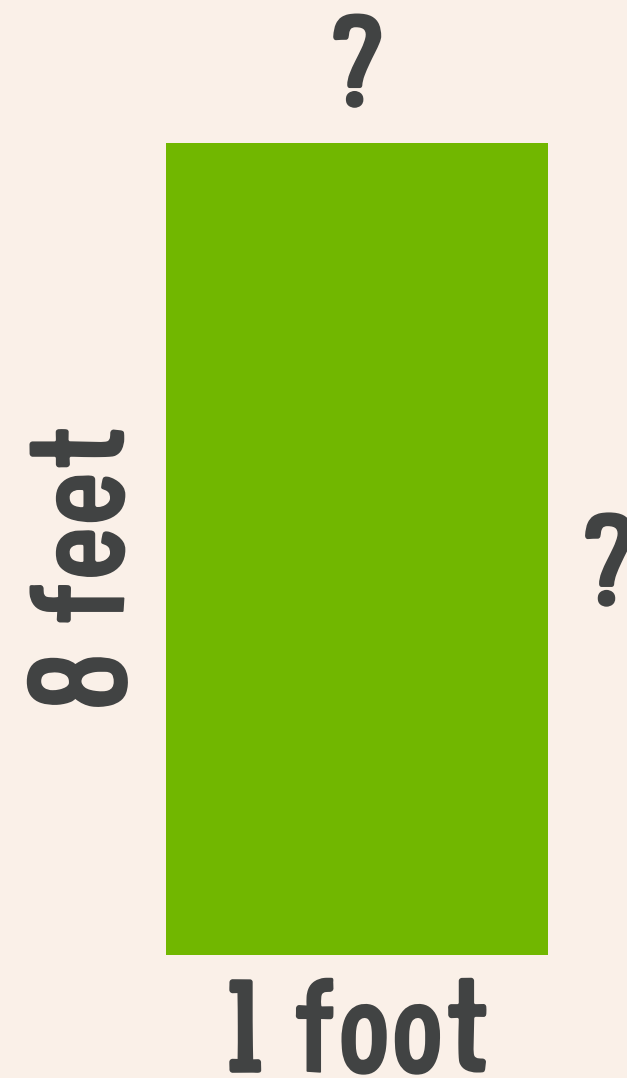
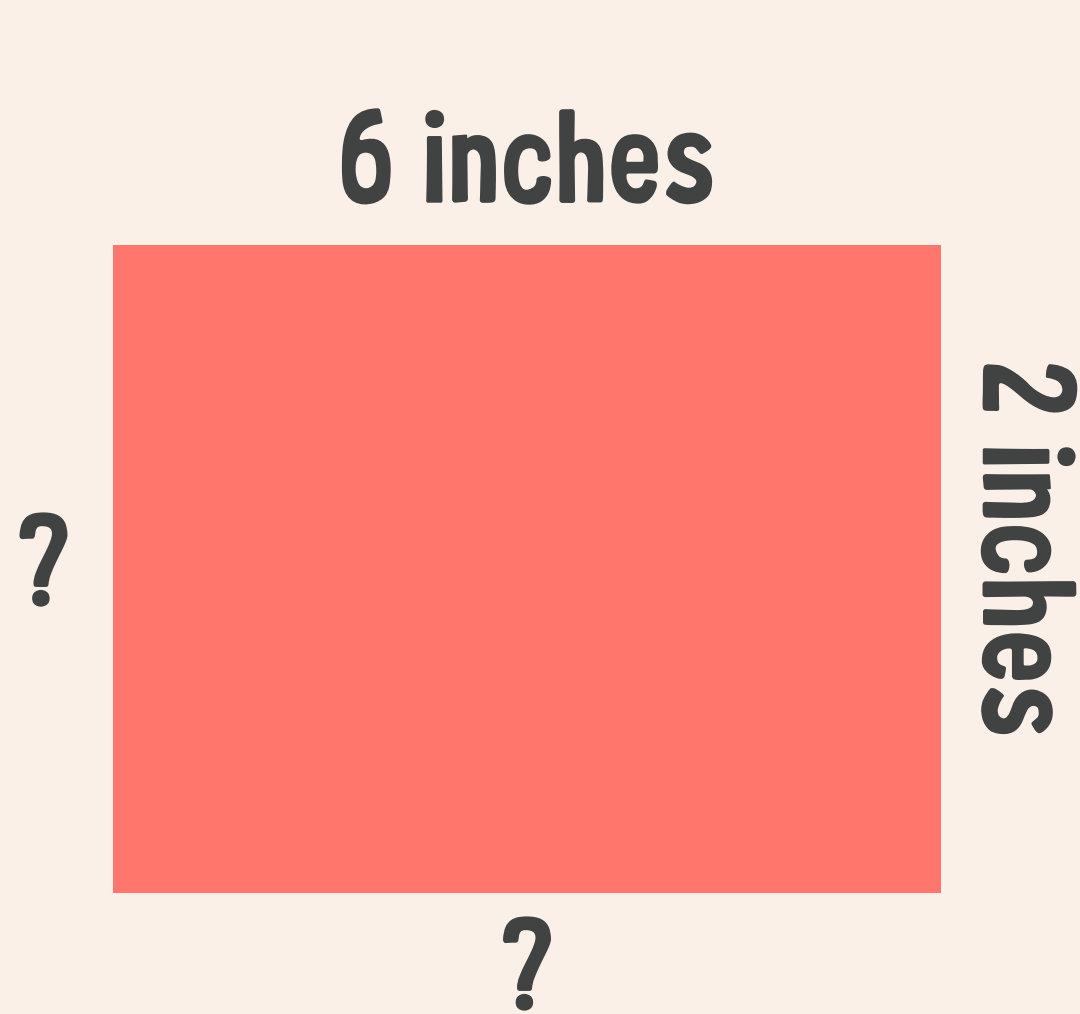


Missing Side Lengths

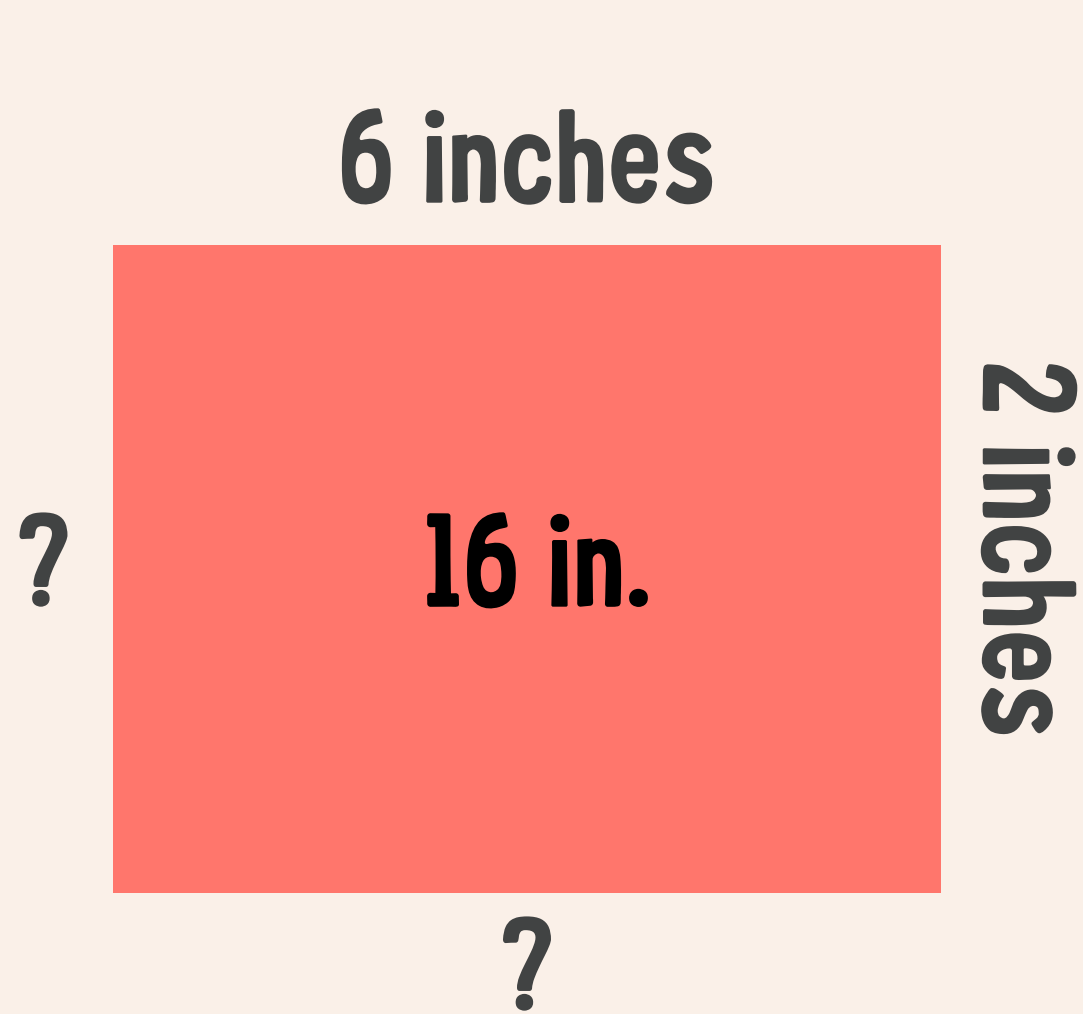
You can find the length of missing sides by looking at parallel sides.



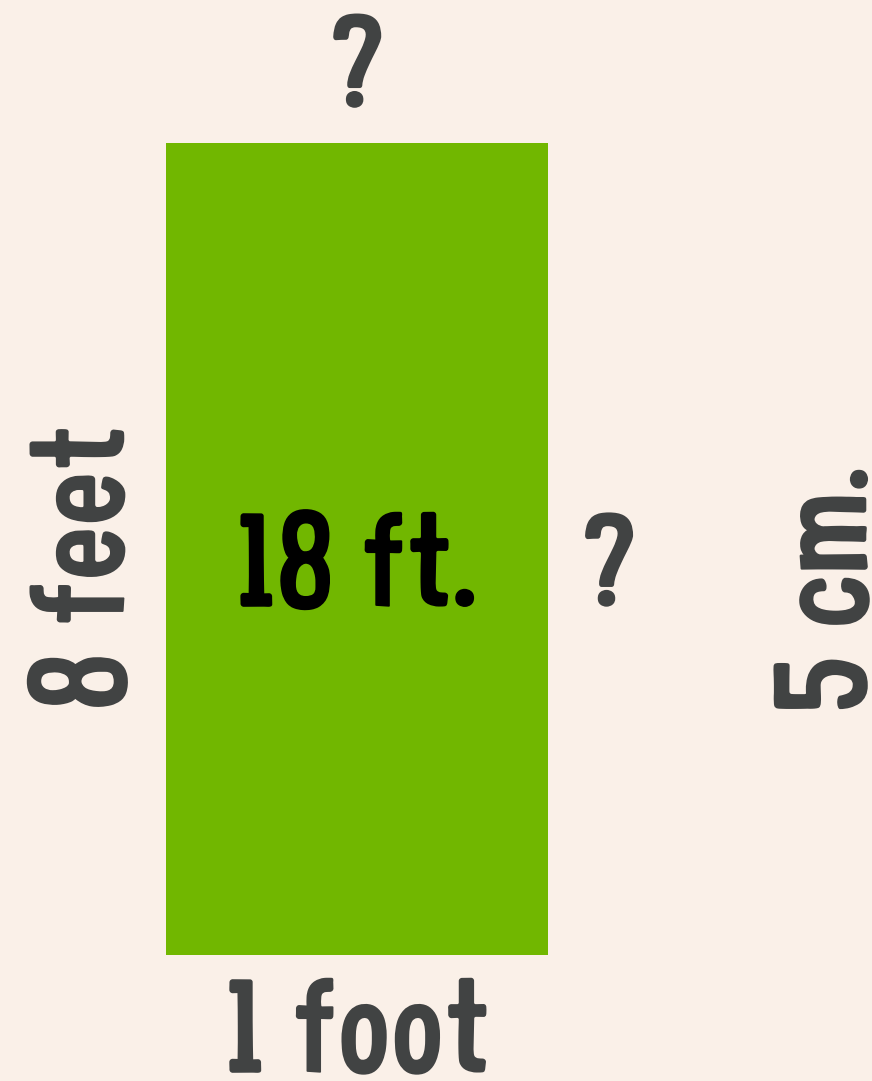
What is the Perimeter of These Figures?



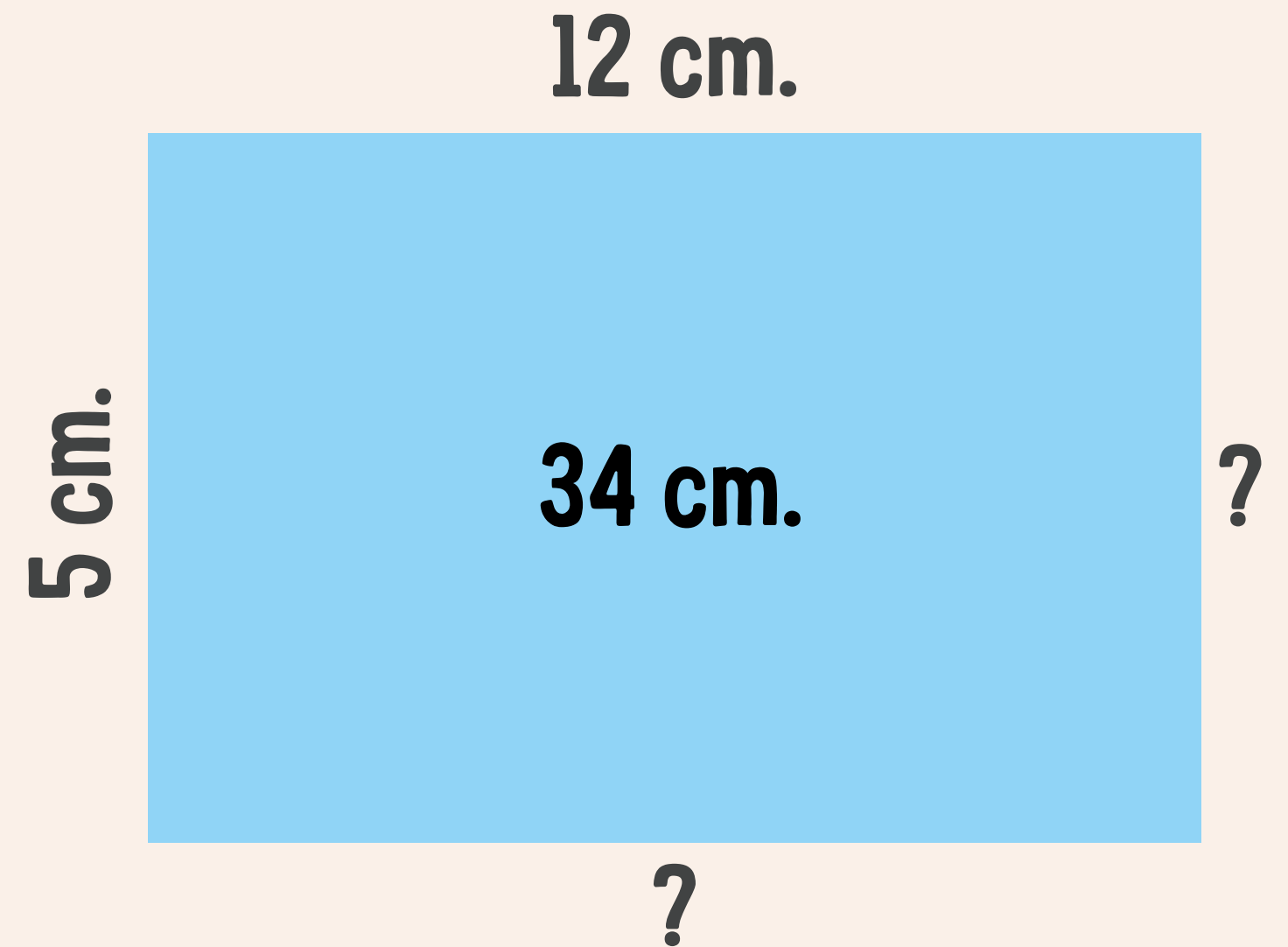
ANSWERS:



$$6 + 6 + 2 + 2 = 16 \text{ in.}$$

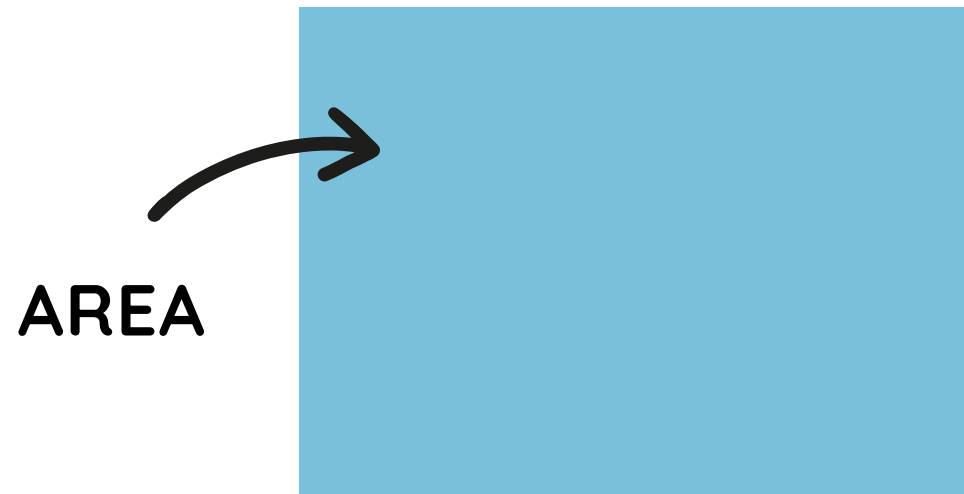
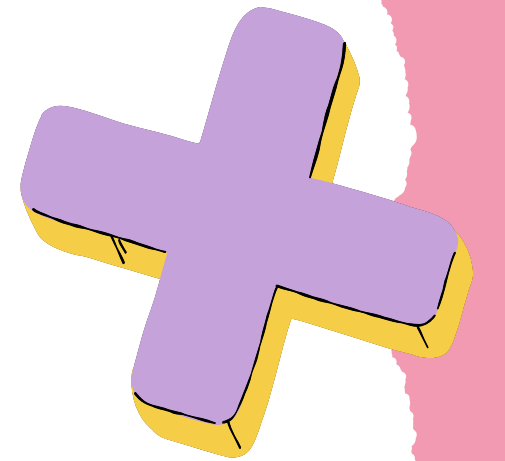


$$8 + 8 + 1 + 1 = 18 \text{ ft.}$$



$$12 + 12 + 5 + 5 = 34 \text{ cm.}$$

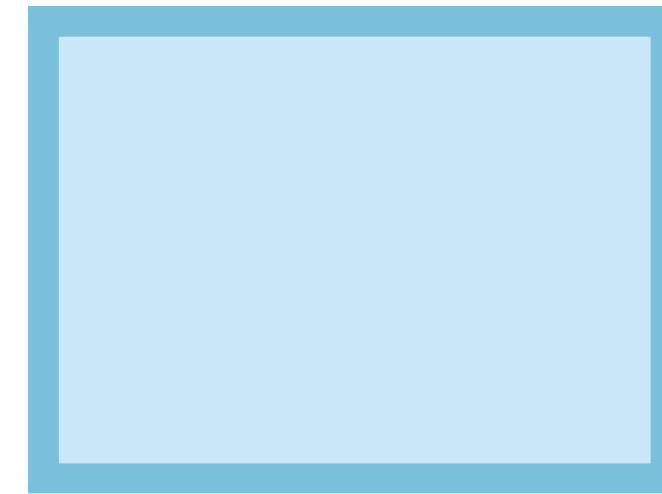
Area vs. Perimeter



AREA

Area

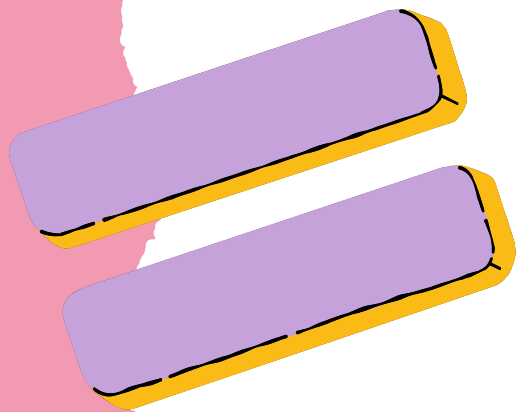
The amount of space
inside a shape.

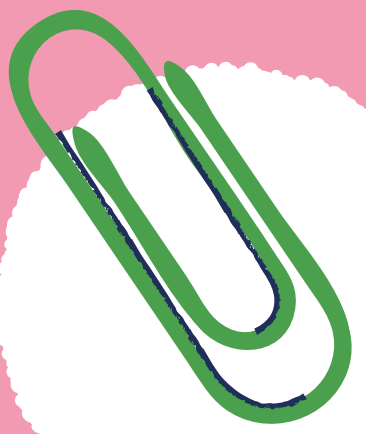


PERIMETER

Perimeter

The total distance around
the outside of a shape.

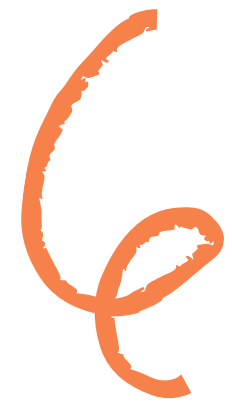
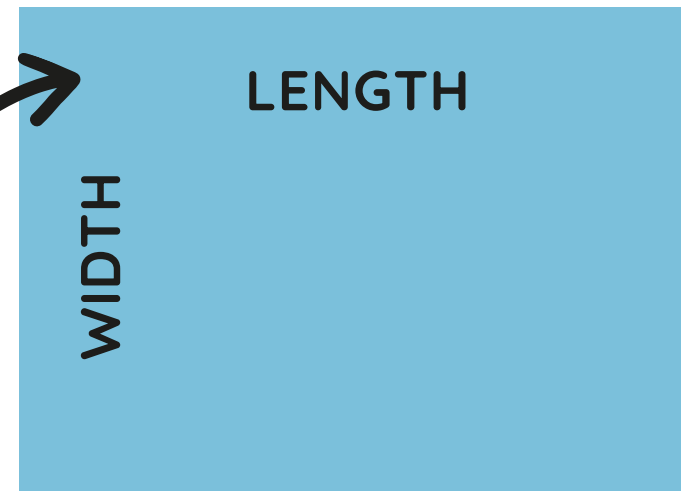




Area vs. Perimeter

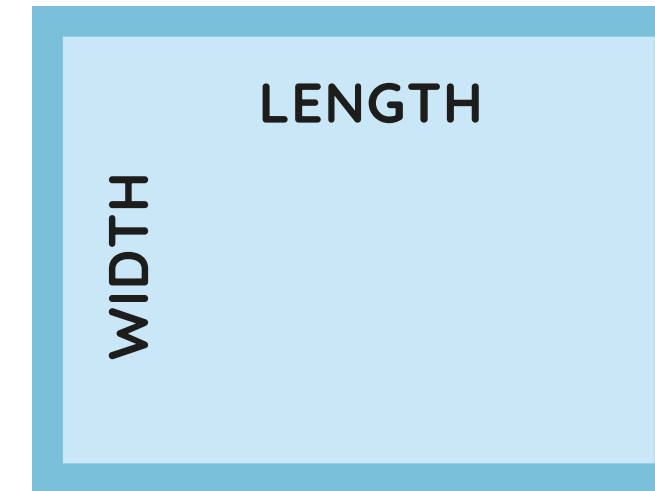


AREA =
LENGTH X
WIDTH



Area

The area is the product of length and width. Area is expressed in square units.



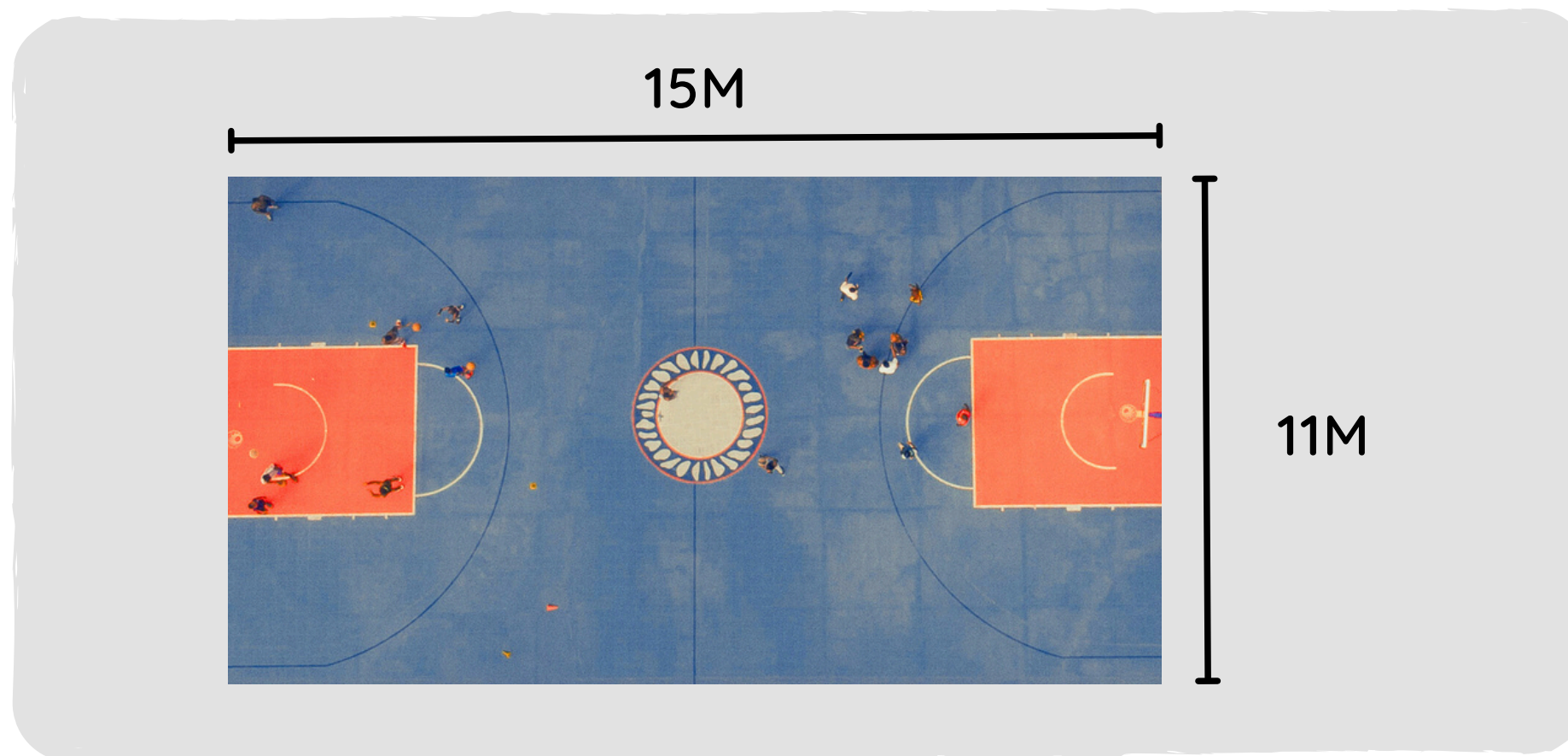
PERIMETER =
2 (LENGTH +
WIDTH)

Perimeter

The perimeter is twice the sum of the length and width. Perimeter is expressed in units.

Example

Let's find the area and perimeter of this basketball court.

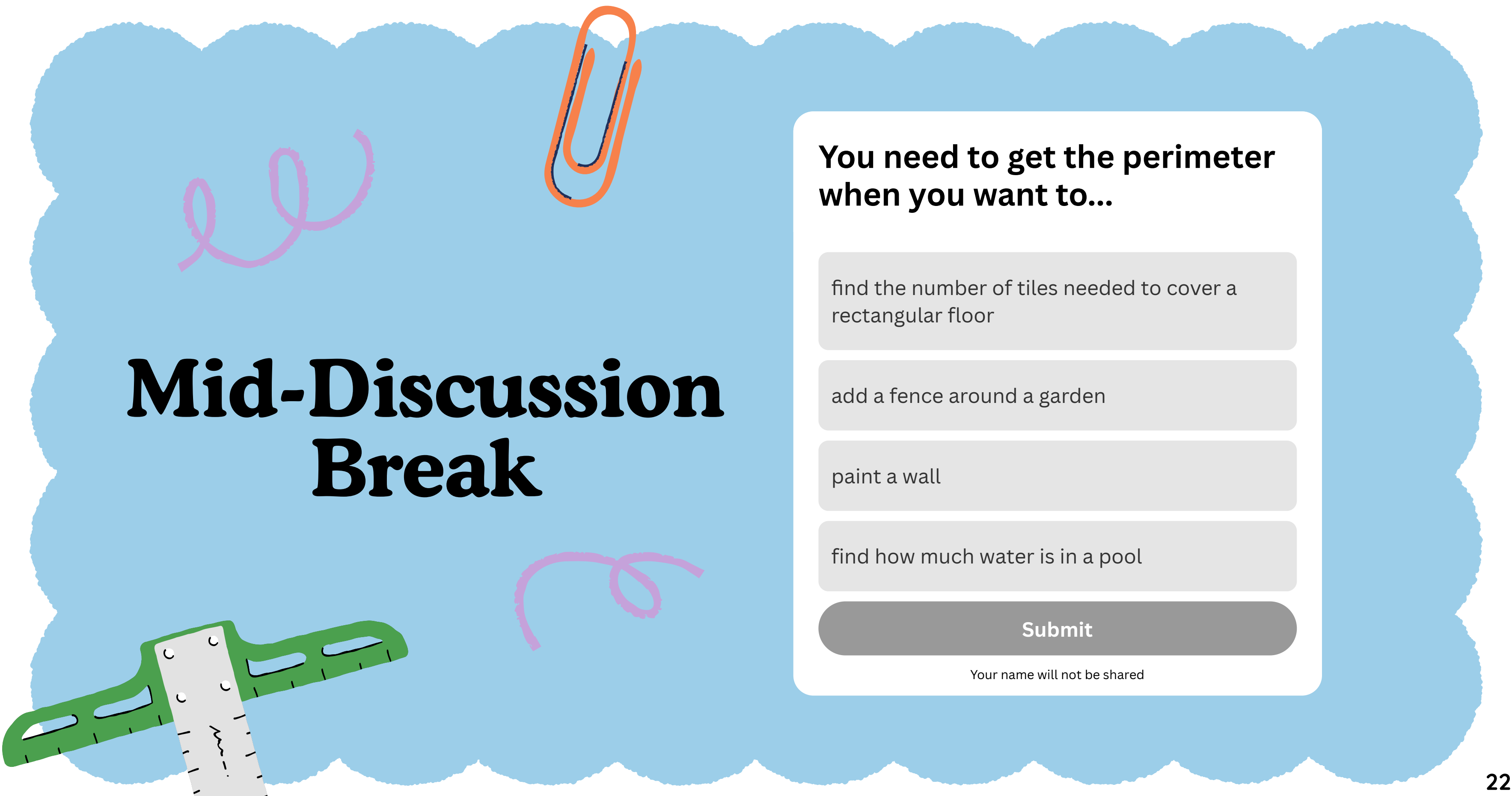


Area

Area = length $\times$ width
Area = 15 m $\times$ 11 m
Area = 165 sq. m

Perimeter

Perimeter = 2 (length + width)
Perimeter = 2 (15 m + 11 m)
Perimeter = 2 (26 m)
Perimeter = 52 m



Mid-Discussion Break

**You need to get the perimeter
when you want to...**

find the number of tiles needed to cover a
rectangular floor

add a fence around a garden

paint a wall

find how much water is in a pool

Submit

Your name will not be shared

Applying Area and Perimeter

1



The rectangular field shown is 310 ft. long and 250 ft. wide. What is its area?

560 ft²

1120 ft²

77,500 ft²

155,000 ft²

Submit

Your name will not be shared

2



The window has a length of 160 cm and a width of 80 cm. What is its perimeter?

480 cm

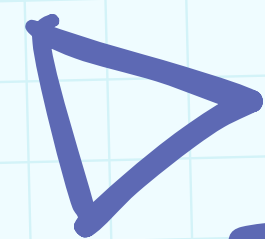
240 cm

320 cm

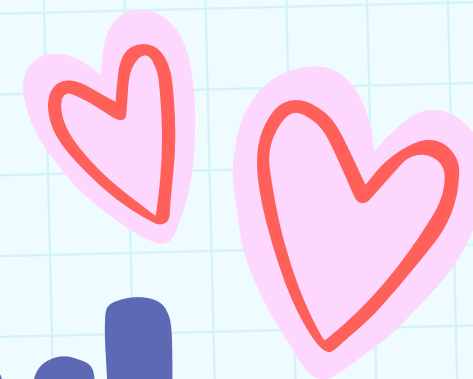
12,800 cm

Submit

Your name will not be shared



Thank You!



To be continued